

Puzzle Theory of the Full 600-Cell

Geometry, Topology, Group Actions, and Sound Optimization

C600 Studio Research Supplement

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1 Purpose and evidence

The full-cut 600-cell makes a useful meeting point between regular-polytope geometry, finite group actions, constructive algorithms, and graphics engineering. Its exterior has only 600 tetrahedral cells, but the retained subdivision has 259,800 sticker slots on 177,120 surface pieces. Consequently, understanding the base polytope is necessary but insufficient for understanding the puzzle. A drawing of the 600-cell, a labelled puzzle configuration, and a camera pose are three different mathematical objects.

This supplement develops that distinction systematically. The geometry determines incidence, local symmetry and the topology of the supporting space. The cuts determine which regions move together. Legal permutations determine reachability and invariants. Ordered local frames make orientation explicit, while complete permutation witnesses connect mathematical controllers to recoverable software operations. The final engineering section explains which optimizations preserve those contracts and which need further measurement or proof.

The principal software contribution remains a modification of **Magic Puzzle Ultimate by Andrey Astrelin**, who receives primary credit for the original simulator and renderer. Nan Ma’s solving methodologies and ivan216’s projects are background to the broader development history, as documented in the companion technical report. They are not attributed the new derivations or the Rethlas review in this supplement.

1.1 Three levels of evidence

Level	What is established	Boundary
Exact base geometry	Coordinate arithmetic, incidence, quaternion closure, explicit symmetries, mod-2 boundary ranks and graph distances	Does not reconstruct the full cut subdivision
Retained puzzle certificates	Orbit census, legal generator maps, local frame groups, parity constraints and selected controller certificates	Conditional inputs to the theorems; not an exact legal-group order computation
Engineering analysis	Consequences for state updates, search, rendering, picking and qualification experiments	Proposals are not claimed as implemented speedups or new GPU benchmarks

The geometry checker uses integer pairs in $\mathbb{Z}[\phi]$, with $\phi^2 = \phi + 1$. It does not infer incidence from floating-point proximity. A separate topological argument identifies the boundary with S^3 ; matching homology ranks alone would not prove that identification. The mathematical blueprint that follows was generated and independently reviewed using Rethlas with a documented Windows I/O adaptation. Its review is an informal AI proof check, not a Lean proof or peer review. The retained artifacts record the actual verdict and scope.

1.2 How the HSC wiki is used

The Hypercubing wiki associated with the Hyperspeedcube community supplies useful definitions and examples. Its **Formal Introduction to Grip Theory** distinguishes solved grips, current active grips and attitude; this helps connect geometric cap membership to piece motion. Its **Piece Invariants** explains how parity and abelian orientation information constrain reachable states. Its **God’s Number** emphasizes a specified move metric when deriving counting bounds, and **Cut Depths** provides normalized cut-distance conventions.

Those pages do not supply this full-cut 600-cell’s 35-orbit census. In particular, the wiki’s A_4 monoflip example belongs to a hypercube setting; the full 600-cell’s twenty-sticker moving orbit has the retained local orientation group A_5 . The relevant wiki pages are linked in the reading guide. Their definitions guide the exposition; the specialized arguments and finite checks are provided here.

The term *complete description* below means a connected account of the specified model and the limits of present knowledge. It does not assert a presentation or exact order for its entire legal twist group, a classification of all cut-depth regimes, or an optimal solver.

1.3 Objects that must remain distinct

Object	Size or type	Role
P	Convex four-dimensional polytope	Geometric body supporting the construction
∂P	Topological three-sphere	Exterior space subdivided into tetrahedra and sticker regions
I^*	120 unit quaternions	Coordinate group underlying the regular vertices
K^+	7,200 elements	Orientation-preserving rigid symmetries of the whole polytope
$K = W(H_4)$	14,400 elements	Full rigid symmetry group, including reflections
$SO(4)$	Continuous group, dimension six	Intended 4D camera rotations
Cap rotation group	A_4 , order 12	Rotations permitted within one tetrahedral cap
H_o	Retained local frame group	Orientations of one piece in moving orbit o
G	Finite; exact order not established here	Group generated by the legal twists on labelled sticker slots

Topological dimension is equally important. A tetrahedral cell of the 600-cell is three-dimensional, and a sticker region in that cell has three-dimensional relative interior. Its displayed mesh uses triangles to draw two-dimensional boundary faces. The application’s rank is a cap-membership statistic and may exceed four; it is not an ambient dimension. A legal permutation of separated sticker regions is also not automatically a continuous self-map of the originally glued boundary. In particular, the trivial fundamental group of S^3 says nothing about the number of legal scrambles or the noncommutativity of twists.

2 Mathematical development

The following blueprint separates unconditional statements about the regular base polytope from implications that explicitly assume the retained puzzle certificates. Products of moves use the stated chronological convention. A geometric symmetry is not silently counted as a free puzzle move.

2.1 Definitions and mathematical sources

Put

$$\phi = \frac{1 + \sqrt{5}}{2}, \quad \tau = \phi^{-1} = \phi - 1. \quad (1)$$

Thus

$$\phi^2 = \phi + 1, \quad \tau^2 = 2 - \phi. \quad (2)$$

Identify $\mathbb{R}^4$ with the quaternion algebra $\mathbb{H}$ using the ordered orthonormal basis $1, i, j, k$. Quaternion conjugation is denoted by $x \mapsto \bar{x}$, and

$$\mathrm{Sp}(1) = \{x \in \mathbb{H} : |x| = 1\}. \quad (3)$$

The binary icosahedral group I^* is the inverse image of the rotation group of a regular icosahedron under

$$\mathrm{Sp}(1) \longrightarrow \mathrm{SO}(3), \quad p \longmapsto (u \mapsto pup^{-1}), \quad (4)$$

where $\mathbb{R}^3$ is identified with the imaginary quaternions.

For a finite set X , write S_X and A_X for its symmetric and alternating groups. Write C_m for the cyclic group of order m , and

$$D_5 = \langle R, F \mid R^5 = F^2 = 1, FRF = R^{-1} \rangle \quad (5)$$

for the dihedral group of order 10.

A flag of a four-dimensional polytope is an incident chain consisting of a vertex, edge, two-face, and facet. A polytope is regular when its geometric symmetry group is transitive on flags.

Two external geometric results will be used, with the following complete statements and conventions.

Quaternion coordinate result. In a suitable orthonormal quaternion coordinate system, the inverse image of the icosahedron’s rotation group consists precisely of

$$(\pm 1, 0, 0, 0), \quad \frac{1}{2}(\pm 1, \pm 1, \pm 1, \pm 1), \quad \frac{1}{2}(\pm \phi, \pm 1, \pm \tau, 0), \quad (6)$$

together with their even coordinate permutations, with independent signs.

Source identifiers: `paper_id = Baez, From the Icosahedron to E8`; `arXiv_id = 1712.06436v2`; `theorem_id = unnumbered coordinate assertion in "The Icosians"`. Here the finite group is the binary lift, not the infinite ring of integral combinations called the icosians. [Baez, “The Icosians”](#).

Binary-group polytope result. The binary icosahedral subgroup of unit quaternions is one orbit of the reflection group of type H_4 , and its elements are the vertices of a regular 600-cell.

Source identifiers: `paper_id = Choi--Lee, Binary Icosahedral Group and 600-Cell, Symmetry 10 (2018), 326`; `theorem_id = Theorem 2`; `arXiv_id = not supplied`. The paper’s binary-group notation indicates a double covering, not coordinate dilation. Its proof establishes reflection invariance in Lemma 2, then identifies the orbit of 1 using its H_3 stabilizer and the 120-element orbit count. Its quaternion norm is the ordinary Euclidean norm used here. [Choi–Lee, Theorem 2 and its proof](#).

These results concern the uncut polytope. They provide no sticker census or puzzle reachability theorem.

2.2 Lemma 1: Coordinate identification

Statement

The set V in the problem has 120 distinct elements, every element has norm 2, and $V/2$ is a binary icosahedral subgroup of $\text{Sp}(1)$. Its convex hull is a regular 600-cell.

Proof

The three coordinate families are disjoint because their patterns of zero coordinates and absolute values differ. Their cardinalities are

$$8, \quad 16, \quad 12 \cdot 8 = 96. \quad (7)$$

In the third family all four absolute values are distinct, so the twelve even permutations give twelve distinct arrangements for each sign choice.

The squared norms are respectively

$$4, \quad 4, \quad 1 + \phi^2 + \tau^2 = 4. \quad (8)$$

The permutation taking

$$(\phi, 1, \tau, 0) \quad \text{to} \quad (0, 1, \phi, \tau) \quad (9)$$

has index sequence $(4, 2, 1, 3)$ and is even. Thus the displayed coordinate set agrees exactly with twice the quaternion coordinate set stated above, including its permutation convention.

The coordinate result identifies $V/2$ with I^* . This identification uses the inverse image of an icosahedral rotation group, hence supplies multiplication closure and inverses, not merely a set of 120 points on a sphere. One direct route to its coordinates is to lift the identity and the rotations about edge, face, and vertex axes using

$$\cos(\theta/2) + u \sin(\theta/2) \quad (10)$$

and its negative, where u is the unit imaginary axis. The rotations comprise the identity, fifteen half-turns, twenty third-turns, and twenty-four fifth-turn rotations.

The binary-group polytope result now applies in the same unit-quaternion metric. Multiplication by 2 preserves regularity and the face lattice. Therefore $P = \text{conv}(V)$ is a regular 600-cell. $\square$

2.3 Lemma 2: Radial topology of the boundary

Statement

The boundary ∂P is homeomorphic to S^3 . Consequently,

$$H_j(\partial P; \mathbb{Z}) \cong \begin{cases} \mathbb{Z}, & j = 0, 3, \\ 0, & j \neq 0, 3, \end{cases} \quad \pi_1(\partial P) = 1. \quad (11)$$

Proof

The polytope contains

$$\text{conv}\{\pm 2e_1, \dots, \pm 2e_4\}, \quad (12)$$

which contains the Euclidean unit ball. Indeed, $\|x\|_2 \leq 1$ implies $\|x\|_1 \leq 2$. Thus 0 lies in the interior of P . The polytope is compact and contained in the ball of radius 2.

For every $u \in S^3$, convexity and compactness give

$$P \cap \{tu : t \geq 0\} = \{tu : 0 \leq t \leq r(u)\} \quad (13)$$

for a unique $r(u) > 0$. Every point tu with $0 \leq t < r(u)$ is interior: it is a convex combination, with positive weight on the first term, of an interior point 0 and the point $r(u)u \in P$. Hence $r(u)u$ is the unique boundary point on this ray.

The map

$$\partial P \longrightarrow S^3, \quad x \longmapsto \frac{x}{\|x\|} \quad (14)$$

is therefore a continuous bijection. Its domain is compact and its codomain Hausdorff, so it is a homeomorphism.

The sphere S^3 has a CW decomposition with one zero-cell and one three-cell, giving the stated integral homology. For simple connectedness, cover S^3 by the complements of two distinct antipodal points. Each complement is homeomorphic to $\mathbb{R}^3$, and their intersection is path connected. The van Kampen theorem gives trivial fundamental group.

This proves the topology directly. Neither mod-2 Betti numbers nor equality of integral homology groups alone would establish the homeomorphism or fundamental group. $\square$

2.4 Lemma 3: Faces, links, metric data, and duality

Statement

The polytope P has circumradius 2, edge length $2/\phi$, and

$$(f_0, f_1, f_2, f_3) = (120, 720, 1200, 600). \quad (15)$$

Its facets are regular tetrahedra, five facets meet at each edge, twenty meet at each vertex, and its vertex links are icosahedral. Its Schläfli symbol is $\{3, 3, 5\}$, and its polar dual is a regular 120-cell of type $\{5, 3, 3\}$.

Proof

Every point of V is an exposed vertex: the functional $x \mapsto v \cdot x$ has its unique maximum at v among points of V , since all have norm 2. Thus P has exactly 120 vertices and circumradius 2.

Let $v = (2, 0, 0, 0)$. The largest first coordinate of another vertex is ϕ , attained at twelve vertices. Consequently the largest inner product with v among distinct vertices is 2ϕ , and the minimum squared distance is

$$8 - 4\phi = 4\tau^2. \quad (16)$$

The group I^* acts transitively on V by left multiplication, so this is also the global minimum distance.

A closest pair of equal-norm vertices is an edge. To see this, let their common squared norm be R^2 and their inner product be $m < R^2$. For any other vertex z , minimality of their distance gives

$$(v + w) \cdot z \leq 2m < R^2 + m = (v + w) \cdot v = (v + w) \cdot w. \quad (17)$$

Thus the functional $v + w$ exposes precisely their segment.

Regularity makes all edges congruent. Hence the twelve closest vertices are exactly the neighbors of v , and the edge length is $2\tau = 2/\phi$.

Their projections onto $v^\perp$ have coordinates

$$(0, \pm 1, \pm \tau) \quad (18)$$

and cyclic permutations. These are a scaled, orthogonally positioned regular icosahedron. All twelve neighboring vertices have first coordinate ϕ , so a sufficiently close hyperplane section at v is similar to this projected set. It follows that the vertex figure, and hence the vertex link, is icosahedral.

An icosahedron has twelve vertices, thirty edges, and twenty triangular faces. These numbers can also be obtained directly from its coordinates: each vertex has five nearest neighbors arranged in a pentagon; the resulting triangular boundary has

$$E = 12 \cdot 5/2 = 30, \quad F = 2E/3 = 20. \quad (19)$$

Its triangular faces correspond to facets through v , giving twenty tetrahedral facets through each vertex. Tetrahedrality also follows from the regular 600-cell identification in Lemma 1.

Double counting yields

$$f_1 = \frac{120 \cdot 12}{2} = 720, \quad f_3 = \frac{120 \cdot 20}{4} = 600. \quad (20)$$

Each tetrahedron has four triangular faces, and each ridge of a convex four-polytope belongs to two facets. Therefore

$$f_2 = \frac{600 \cdot 4}{2} = 1200. \quad (21)$$

The five triangles incident to a vertex of an icosahedral link correspond to the five facets incident to the associated edge of P . Thus the Schläfli symbol is $\{3, 3, 5\}$.

The Euler characteristic is

$$120 - 720 + 1200 - 600 = 0, \quad (22)$$

in agreement with Lemma 2.

Because 0 is interior, the polar

$$P^\circ = \{y : y \cdot x \leq 1 \text{ for every } x \in P\} \quad (23)$$

is a bounded four-polytope. Polarity reverses proper face incidences and sends a j -face to a $(3 - j)$ -face. Orthogonal symmetries commute with polarity, so the dual is regular. Its facets are dual to the icosahedral vertex figures and hence are dodecahedra. It has

$$(f_0, f_1, f_2, f_3) = (600, 1200, 720, 120) \quad (24)$$

and reversed symbol $\{5, 3, 3\}$: it is the regular 120-cell.

This duality is not an identification of the two polytopes or their face counts. $\square$

2.5 Lemma 4: The finite geometric symmetry group

Statement

For $p, q \in I^*$, define

$$L_{p,q}(x) = pxq^{-1}. \quad (25)$$

Then

$$K^+ \cong (I^* \times I^*) / \langle (-1, -1) \rangle, \quad |K^+| = 7200, \quad (26)$$

and quaternion conjugation extends this to K of order 14400.

Moreover,

$$K \cong W(H_4) \quad (27)$$

with presentation

$$\left\langle s_0, s_1, s_2, s_3 \left| \begin{array}{l} s_i^2 = 1, \\ (s_0 s_1)^3 = (s_1 s_2)^3 = (s_2 s_3)^5 = 1, \\ (s_i s_j)^2 = 1 \quad \text{if } |i - j| > 1 \end{array} \right. \right\rangle. \quad (28)$$

Proof

Quaternion norm multiplicativity makes $L_{p,q}$ orthogonal. Its determinant is positive: the same construction on the connected space $\mathrm{Sp}(1) \times \mathrm{Sp}(1)$ depends continuously on (p, q) and includes the identity. Multiplication closure of I^* shows that $L_{p,q}$ preserves $V/2$, and therefore P .

If $L_{p,q}$ is the identity, evaluating at 1 gives $p = q$. This quaternion must then commute with every quaternion. Commuting with i and j forces it to be real, so it is 1 or -1 . The kernel is precisely

$$\{(1, 1), (-1, -1)\}. \quad (29)$$

Thus these maps give $120^2/2 = 7200$ distinct positive-determinant symmetries.

Quaternion conjugation

$$J(x) = \bar{x} \quad (30)$$

preserves V , since signs are independent. Its real matrix is

$$\mathrm{diag}(1, -1, -1, -1), \quad (31)$$

so $\det J = -1$. The maps $L_{p,q}$ and $L_{p,q}J$ therefore give 14400 distinct symmetries.

An upper bound is still necessary. Fix a vertex v . Its stabilizer acts faithfully on the icosahedral vertex figure: the vertex direction and the three-dimensional span of that figure together span $\mathbb{R}^4$. An orthogonal symmetry of an icosahedron is determined by the image of one ordered triangular face. There are at most

$$20 \cdot 6 = 120 \quad (32)$$

such images. Orbit–stabilizer consequently gives

$$|K| \leq 120 \cdot 120 = 14400. \quad (33)$$

The exhibited maps attain this upper bound, proving both group orders and the description of K^+ .

Also,

$$JL_{p,q}J = L_{q,p}, \quad (34)$$

so conjugation exchanges the two quaternion factors.

For the Coxeter presentation, radially project the barycentric subdivision of ∂P onto S^3 . A flag gives a spherical tetrahedral chamber. A symmetry fixing a flag fixes the barycenters of its four faces; these four vectors are linearly independent, so the symmetry is the identity. Regularity therefore gives a simply transitive action on chambers.

For each of the four chamber walls, the symmetry taking the chosen chamber to its adjacent chamber fixes the other three barycenter rays. It is the reflection in their span. Denote these reflections by $s_0, \dots, s_3$, indexed by face ranks.

Chamber adjacency is connected, so these reflections generate K . Consecutive rank changes give rank-two sections of sizes 3, 3, 5, producing the displayed relations. Nonconsecutive changes commute and give square galleries.

These relations are complete. A word acting trivially determines a closed gallery in the chamber complex. The dual cell decomposition is a cell decomposition of S^3 , which is simply connected by Lemma 2. Its two-skeleton therefore has trivial fundamental group. Closed galleries are generated by backtracking and boundaries of dual two-cells. Those boundaries are precisely the involution, square, and rank-two polygon relations above. Hence the displayed presentation presents K , rather than merely mapping onto it.

The Coxeter group with this chain presentation is $W(H_4)$. $\square$

2.6 The finite checker and its scope

The supplied checker implements a finite audit in the exact ring

$$\mathbb{Z}[\phi] = \{a + b\phi : a, b \in \mathbb{Z}\}. \quad (35)$$

It represents an element by (a, b) and uses

$$(a + b\phi)(c + d\phi) = (ac + bd) + (ad + bc + bd)\phi. \quad (36)$$

Its sign test compares integers after writing

$$2(a + b\phi) = (2a + b) + b\sqrt{5}, \quad (37)$$

with sign cases handled before squaring. Thus its incidence tests do not depend on floating-point tolerances.

Its algorithm:

1. Enumerates the three vertex families and verifies norms.
2. Joins pairs with inner product 2ϕ .
3. Enumerates graph triangles and four-cliques.
4. For each four-clique, forms the sum N of its vertices and checks a strict supporting inequality at every other vertex.
5. Counts incidences and constructs vertex links.
6. Computes boundary matrices and ranks over $\mathbb{F}_2$.
7. Tests quaternion multiplication closure and counts distinct linear maps by their images on a basis.

For a tetrahedron with vertices $v_1, \dots, v_4$, the Gram matrix has diagonal entries 4 and off-diagonal entries 2ϕ . It is positive definite. Moreover,

$$N = \sum_i v_i, \quad N \cdot v_i = 4 + 6\phi, \quad \|N\|^2 = 16 + 24\phi. \quad (38)$$

Thus the supporting tests certify actual tetrahedral facets, not merely graph cliques.

There is also a completeness argument beyond finding some supporting facets. If every triangular ridge of the listed tetrahedra has both its incident facets in the list, the list is closed under facet adjacency. The facet adjacency graph of a convex polytope is connected, so a nonempty such list contains all facets.

Nevertheless:

- A link's f -vector alone does not identify it as an icosahedron; Lemma 3 supplies its geometry.
- Ranks over $\mathbb{F}_2$ do not prove integral homology or simple connectedness; Lemma 2 supplies those conclusions.
- Exhibiting 14400 maps does not exclude further symmetries; Lemma 4 supplies the upper bound.
- No part of this checker reconstructs the cut subdivision, verifies all legal sticker maps, or determines the full legal group.

The supplied JSON records a previous reported run. The unconditional proofs above do not take its `passed` field as a substitute for these arguments.

2.7 Lemma 5: Cell-pole normalization

Statement

The radius-two model has inradius

$$r_{\text{in}} = \frac{\phi^2}{\sqrt{2}}. \quad (39)$$

A similarity with scale factor $4/\phi$ and a suitable rotation places a chosen facet pole at

$$n_0 = (\phi^3, 1, 1, 1), \quad (40)$$

with every facet plane written as

$$n_c \cdot x = \|n_c\|^2. \quad (41)$$

Proof

For a tetrahedral facet, use N from the preceding calculation. Since

$$\|N\|^2 = 4(4 + 6\phi), \quad (42)$$

the perpendicular foot of the origin on its plane is

$$n = \frac{N}{4}. \quad (43)$$

Its squared norm is

$$\|n\|^2 = \frac{4 + 6\phi}{4} = \frac{\phi^4}{2}. \quad (44)$$

Regularity gives the same inradius at every facet.

On the other hand,

$$\|n_0\|^2 = \phi^6 + 3 = 8\phi^2. \quad (45)$$

Therefore

$$\frac{\|n_0\|}{r_{\text{in}}} = \frac{4}{\phi}. \quad (46)$$

Choose $Q \in SO(4)$ taking the direction of one original pole to that of n_0 , and put

$$\tilde{P} = \frac{4}{\phi}QP. \quad (47)$$

Every original pole n becomes $\tilde{n} = (4/\phi)Qn$, and its plane becomes

$$\tilde{n} \cdot x = \|\tilde{n}\|^2. \quad (48)$$

The selected pole is exactly n_0 .

All subsequent puzzle geometry uses $\tilde{P}$, identified with P by this fixed similarity. Its geometric symmetry group is the conjugate of K under the similarity. The radius-two and cell-pole coordinate scales are not numerically identified. $\square$

2.8 Retained-model definitions and hypotheses

Let $\mathcal{C}$ be the set of 600 facets, and let n_c denote their poles in the normalization of Lemma 5. Put

$$\alpha = \frac{121}{125}, \quad U_c = \{x \in \tilde{P} : n_c \cdot x > \alpha\|n_c\|^2\}. \quad (49)$$

Central symmetry pairs each n_c with $-n_c$. Thus there are 600 oriented caps or, equivalently, 300 antipodal axes. For one axis, the two outer layers satisfy

$$n_c \cdot x > \alpha\|n_c\|^2 \quad \text{and} \quad n_c \cdot x < -\alpha\|n_c\|^2. \quad (50)$$

The middle layer lies between them and is fixed during that axis's outer-layer turns. It is not asserted to be fixed under turns about every other axis.

A full-dimensional cut chamber is specified by choosing a strict side of every cut hyperplane and intersecting these choices with $\text{int } \tilde{P}$. A nonempty such intersection is convex and hence connected. A surface piece is a retained chamber whose closure meets the boundary in one or more relatively three-dimensional patches. Cut-boundary strata are attached consistently to these pieces; they are not additional sticker slots.

For a piece position A , define its complete cap signature

$$M(A) = \{c : \text{the chamber interior of } A \text{ lies in } U_c\}. \quad (51)$$

Define its hosting set by

$$\text{Host}(A) = \{h : \bar{A} \cap h \text{ contains a retained relatively three-dimensional patch}\}. \quad (52)$$

A sticker slot is such a patch. In this full-cut description it is addressed by

$$(M(A), h), \quad h \in \text{Host}(A). \quad (53)$$

The following hypotheses specify exactly the retained evidence used below.

R — Retained subdivision and legal action. The retained model is the complete surface subdivision just described, with no hidden identification of distinct pieces having the same recorded address. Legal cap rotations carry whole pieces and their sticker patches to whole pieces and patches. It has 259800 sticker slots, 177120 surface pieces, 600 one-sticker centers fixed pointwise by every legal move, and the following 35 moving orbits. The frame groups are faithful groups of permutations of the stickers of a piece after coherent transport.

Orbit	Rank	Stickers per piece	Pieces N_o	Frame group H_o
O00	1	2	1200	C_2
O01	2	1	3600	1
O02	3	1	2400	1
O03	3	2	3600	C_2
O04	4	1	7200	1
O05	4	1	7200	1
O06	4	5	720	D_5
O07	5	1	7200	1
O08	5	2	3600	C_2
O09	6	2	7200	1
O10	7	1	7200	1
O11	7	2	3600	C_2
O12	8	1	7200	1
O13	8	1	7200	1
O14	8	2	7200	1
O15	9	1	2400	1
O16	9	2	7200	1
O17	9	5	1440	C_5
O18	10	1	7200	1
O19	10	1	7200	1
O20	10	2	7200	1
O21	11	1	7200	1
O22	11	2	3600	C_2
O23	12	2	7200	1
O24	13	1	7200	1
O25	13	2	3600	C_2
O26	14	1	7200	1
O27	14	1	7200	1
O28	14	5	1440	C_5
O29	15	1	2400	1
O30	15	2	7200	1
O31	16	1	7200	1
O32	17	2	3600	C_2
O33	18	1	2400	1
O34	19	20	120	A_5

These data satisfy

$$\sum_o N_o = 176520, \quad 600 + \sum_o k_o N_o = 259800. \quad (54)$$

The 600 fixed centers are 600 singleton position orbits, not another moving orbit.

F — Full local cap action. Every cap admits its full tetrahedral rotation group A_4 , acting on the same complete subdivision. The chosen H_c is a nonidentity half-turn and T_c is a third-turn in that action.

I — Retained generator-invariant certificate. In the transported coordinates constructed below, each chosen primitive has even positional permutation separately in every moving orbit and zero total orientation image in each orbit’s abelianization. This is a hypothesis about every generator, not an inference from sample scrambles.

C — Nine pure-controller certificates. For

$$\mathcal{I} = \{1, 4, 5, 7, 18, 19, 27, 31, 33\}, \quad (55)$$

each orbit has a legal word whose complete sticker permutation is a single three-cycle in that orbit and fixes every other sticker. A verified legal relocation produces a cycle $(A_o B_o C_o)$. For every other position X in that orbit, a verified legal setup fixes the two buffer sticker slots A_o, B_o pointwise and sends C_o to X . Coverage means valid transitions and coverage of every nonbuffer destination, not merely a reported search-node count.

The solved coloring has 600 distinct home-cell symbols, with six copies of each in O01, four in O33, and twelve in each of the other seven selected orbits.

E — Equivariant geometric realization. For the extended symmetry action, the entire retained subdivision and its piece grouping are preserved by the geometric action of K . Each center slot is geometrically distinguished at its corresponding pole n_c , and geometric permutations act on the same labelled sticker set as the legal moves. In particular, this is a faithful geometric realization, not an action on a collapsed display quotient.

Hypotheses R, F, I, C, and E have different roles. In particular, C does not certify I, and neither certifies a reconstruction of all cuts. The supplied arithmetic and controller summaries are retained evidence for these hypotheses; their reported pass fields do not replace their full hypotheses.

No independent derivation of the 433 retained regions per tetrahedron is claimed.

2.9 Lemma 6: Cap actions, signatures, and piece equivariance

Statement

Under R and F, H_c, T_c generate the full A_4 action at each cap. The sticker-to-piece map

$$\rho : \mathcal{S} \longrightarrow \mathcal{P} \quad (56)$$

is equivariant under every legal move and hence every legal word:

$$\rho(g(s)) = \bar{g}(\rho(s)). \quad (57)$$

Complete cap signatures identify the cut-chamber positions in this model, and

$$r(A) = |M(A)| - 1, \quad k(A) = |\text{Host}(A)|. \quad (58)$$

The quantity $r(A)$ is neither a topological dimension nor a codimension.

Proof

The three nonidentity half-turns of a tetrahedron form the nonidentity elements of a normal Klein four subgroup of A_4 . Conjugation by any third-turn cycles these three elements. Thus H_c and its conjugates by T_c generate the Klein four group, and adjoining T_c gives all twelve elements of A_4 .

All stickers of a physical piece travel together under a legal rigid cap rotation by R . Therefore the image piece $\bar{g}(A)$ is well defined independently of the choice of sticker in A , and the equivariance equation follows. The equation is preserved under composition and inversion.

A full cap signature specifies every strict side choice in the cut arrangement. The corresponding chamber is an intersection of open half-spaces with $\text{int } \tilde{P}$, so it has at most one connected component. Thus two different chamber positions cannot have the same complete signature. A hosting patch is an intersection of the chamber closure with a facet, with its lower-dimensional boundaries removed; the cut construction gives at most one such patch per hosting facet.

Consequently $(M(A), h)$ addresses a sticker slot and ρ forgets h . This justification depends on the complete cut-chamber model. In an arbitrary grip-set representation, signature injectivity would require a separate hypothesis.

For a global geometric symmetry k , cap and host labels are permuted:

$$M(kA) = kM(A), \quad \text{Host}(kA) = k \text{Host}(A). \quad (59)$$

A local twist need not act on every piece signature by one common permutation of all cap labels; the destination signature is determined by the actual destination chamber.

The two numerical formulas are the definitions of application rank and sticker multiplicity. They count memberships and hosting patches. Physical cut chambers have four-dimensional interiors, and sticker interiors are three-dimensional, while the retained ranks range up to 19. Thus rank cannot be either dimension or codimension in the ambient four-space.

Nor does rank identify a move orbit: for example, O04 and O05 have the same rank and sticker multiplicity but are distinct retained orbits. Twenty corner stickers also do not imply a cyclic orientation variable of order twenty; the specified corner frame group is A_5 . $\square$

2.10 Lemma 7: Transported frames and the ambient wreath product

Statement

Let

$$G = \langle H_c, T_c : c \in \mathcal{C} \rangle \leq S_S. \quad (60)$$

Under R , coherent transported frame choices give an injective homomorphism

$$G \hookrightarrow \prod_{o=0}^{34} (H_o^{N_o} \rtimes S_{N_o}). \quad (61)$$

With chronological multiplication, the coordinates satisfy

$$(p, a)(r, b) = (r \circ p, c), \quad c_i = a_i b_{p(i)}. \quad (62)$$

Proof

Fix one moving orbit. Choose a reference position and a reference ordering of its stickers. Choose a legal transporter from the reference position to every other position. These transporters identify all sticker fibers with a common reference fiber.

For any legal map taking position i to position j , composing with the chosen transporters gives a return map on the reference fiber. Its faithful permutation action belongs to the retained local frame group H_o .

This constructs coherent coordinates; arbitrary independent screen orderings would not automatically have this property.

Write the position permutation as p , and the frame increment at source position i as $a_i \in H_o$. In the right-action convention, an abstract frame state transforms by

$$(i, h) \mapsto (p(i), ha_i). \quad (63)$$

Applying (p, a) and then (r, b) gives

$$(i, h) \mapsto (r(p(i)), ha_i b_{p(i)}). \quad (64)$$

This proves the multiplication law, including its reindexing.

Every legal move preserves each of its piece orbits, so the construction applies independently to the 35 moving orbits. If all resulting coordinates are trivial, every moving sticker is fixed, because each fiber action is faithful. The center stickers are fixed by R . Hence the original sticker permutation is the identity, proving injectivity.

Changing the coherent reference frames conjugates the coordinate realization; it does not enlarge the legal group.

For a source-to-destination sticker permutation g , a labelled state array obeys

$$L'[g(s)] = L[s]. \quad (65)$$

This is consistent with the chronological convention used throughout. $\square$

2.11 Lemma 8: Necessary invariants and the conditional upper bound

Statement

Under R and I ,

$$|G| \leq \frac{\prod_{o=0}^{34} N_o! |H_o|^{N_o}}{2^{43} 5^2}. \quad (66)$$

The denominator is the exact index of the stated invariant kernel in the ambient product. Equality with $|G|$ is not asserted.

Proof

Let

$$H_o^{\text{ab}} = H_o / [H_o, H_o], \quad \chi_o : H_o \rightarrow H_o^{\text{ab}}. \quad (67)$$

Write the abelian quotient additively and define

$$\Phi_o(p, a) = \sum_i \chi_o(a_i). \quad (68)$$

Lemma 7 gives

$$\begin{aligned} \Phi_o((p, a)(r, b)) &= \sum_i \chi_o(a_i b_{p(i)}) \\ &= \sum_i \chi_o(a_i) + \sum_i \chi_o(b_{p(i)}) \\ &= \Phi_o(p, a) + \Phi_o(r, b). \end{aligned} \quad (69)$$

Thus Φ_o is a homomorphism. Positional sign is also a homomorphism because

$$\text{sgn}(r \circ p) = \text{sgn}(r) \text{sgn}(p). \quad (70)$$

Hypothesis I says that every generator lies in the kernels of these maps, so every legal word does.

The local abelianizations are

$$C_2^{\text{ab}} = C_2, \quad C_5^{\text{ab}} = C_5, \quad D_5^{\text{ab}} = C_2, \quad A_5^{\text{ab}} = 1. \quad (71)$$

For D_5 , the quotient by $\langle R \rangle$ is C_2 , while

$$[R, F] = R^2 \quad (72)$$

generates $\langle R \rangle$. Thus $[D_5, D_5] = \langle R \rangle$.

For completeness, A_5 is perfect: take

$$u = (1\ 2)(4\ 5), \quad v = (2\ 3)(4\ 5). \quad (73)$$

Their commutator is a three-cycle. The derived subgroup is normal, and all three-cycles are conjugate within A_5 . Since three-cycles generate A_5 , its derived subgroup is all of A_5 .

There are thirty-five positional sign constraints, seven C_2 orientation constraints, one D_5 orientation constraint, and two C_5 constraints.

These constraints are independent in the ambient product. Positional parity can be prescribed using a transposition and identity frame increments. Each orientation quotient value can be prescribed using one frame coordinate and the identity positional permutation. Choices in different orbits are independent. Therefore the joint invariant map is onto a group of order

$$2^{35} \cdot 2^7 \cdot 2 \cdot 5^2 = 2^{43} 5^2. \quad (74)$$

Its kernel has precisely this index. Lemma 7 and hypothesis I place G inside that kernel, proving the bound.

A reflection-type element of the natural five-point D_5 action exchanges two pairs and is even as a sticker permutation. Its abelianization bit therefore cannot be replaced by ordinary sticker sign. Likewise, an odd positional permutation in one orbit cannot cancel an odd positional permutation in another under I.

The bound leaves open additional restrictions and reachability sufficiency. $\square$

2.12 Lemma 9: Pure stars generate independent alternating actions

Statement

Under R and C, G contains a subgroup acting as

$$A_{3600} \times A_{2400} \times A_{7200}^7 \quad (75)$$

on the nine selected single-sticker orbits and fixing every other sticker.

Proof

Words execute from left to right. Let

$$t_C = (A\ B\ C) \quad (76)$$

be a relocated pure controller. If S fixes A, B and sends C to X , then the chronological word

$$S^{-1} t_C S \quad (77)$$

has ordinary functional action $S \circ t_C \circ S^{-1}$, so it is exactly

$$t_X = (A\ B\ X). \quad (78)$$

Conjugating the complete pure permutation gives a complete pure permutation. Temporary collateral during its legal witness is irrelevant to this net support statement.

For distinct nonbuffer positions X, Y , direct evaluation gives

$$t_Y^{-1}t_X = (A Y X). \quad (79)$$

Together with the original stars and their inverses, these supply every three-cycle containing A . For distinct $x, y, z \neq A$,

$$(A x y)(A z x) = (x y z). \quad (80)$$

Thus the stars generate every three-cycle.

Every even permutation is a product of three-cycles. For example, express it as an even number of transpositions and pair them; a pair sharing a point is a three-cycle, identical transpositions cancel, and a disjoint pair can be rewritten as

$$(a b)(c d) = (a b c)(a d c) \quad (81)$$

in the chronological convention. Conversely, every star is even. The supported subgroup generated on this orbit is therefore exactly A_N .

The nine supported subgroups have disjoint full sticker supports. They commute, and the multiplication map from their direct product is injective: restricting an identity product to one selected orbit makes that factor the identity. This proves the claim.

This identifies a supported subgroup of G . It does not identify the full legal group or its actions on the other twenty-six moving orbits. $\square$

2.13 Lemma 10: Balanced-color lower bound

Statement

Under R and C, the set Ω of reachable fixed-frame face-color states satisfies

$$|\Omega| \geq B, \quad (82)$$

where

$$B = \frac{3600!}{(6!)^{600}} \frac{2400!}{(4!)^{600}} \left[\frac{7200!}{(12!)^{600}} \right]^7. \quad (83)$$

These states can be attained with every unselected orbit solved at macro boundaries.

Proof

Consider one selected orbit with $N = 600m$ positions and m copies of each color. The stabilizer of its solved coloring in S_N is

$$Q = (S_m)^{600}. \quad (84)$$

Since $m \in \{4, 6, 12\}$, this stabilizer contains a transposition of two equal-colored positions and hence an odd permutation. Therefore

$$|A_N \cap Q| = \frac{|Q|}{2}. \quad (85)$$

The number of colorings in the A_N orbit is

$$\frac{|A_N|}{|A_N \cap Q|} = \frac{N!/2}{(m!)^{600}/2} = \frac{N!}{(m!)^{600}}. \quad (86)$$

The odd same-color transposition is used only to describe the coloring stabilizer. It need not itself be a legal move.

Lemma 9 gives independent alternating actions with full purity, so the nine factors multiply and every unselected sticker remains solved at the ends of the resulting macros. This gives B .

A labelled state distinguishes every home sticker identity and is determined by an element of G . A face-color state forgets distinctions between stickers with the same home-cell symbol. Thus B is a lower bound for colored states, whereas Lemma 8 is an upper bound for labelled states.

The colors here are 600 distinct mathematical symbols in a fixed spatial frame. Repeated display RGB values or additional quotienting by spatial and symbol symmetries would define a different state set and require a separate count. $\square$

2.14 Lemma 11: Word-ball and total-variation bounds

Statement

Under R and C , use the unit-cost alphabet

$$\mathcal{A} = \{H_c, T_c, T_c^{-1} : c \in \mathcal{C}\}, \quad q = 1800. \quad (87)$$

Some reachable face-color state has distance at least 46648 from solved.

If μ_{1000} is any distribution produced by 1000 successive letters of this alphabet from solved and ν is uniform on Ω , then

$$\|\mu_{1000} - \nu\|_{\text{TV}} > 1 - 10^{-148595}. \quad (88)$$

Proof

At most q^j words have length j . Different words may give the same state, so

$$|\{x : d(\text{solved}, x) \leq d\}| \leq \sum_{j=0}^d q^j = \frac{q^{d+1} - 1}{q - 1}. \quad (89)$$

The numerical comparisons needed here are the exact integer inequalities

$$1800^{46648} - 1 < 1799B, \quad (90)$$

and

$$1800^{1000} 10^{148595} < B. \quad (91)$$

They can be evaluated without logarithms: form the factorials in B as integer products, perform the indicated exact multinomial divisions, and compare positive integers. These comparisons are confirmed by the accompanying exact-integer audit. As an arithmetic cross-check,

$$1800^{46649} - 1 \geq 1799B. \quad (92)$$

The cross-check says only when this counting argument stops excluding coverage.

By Equation~(90), the radius-46647 ball contains fewer than B states. Lemma 10 gives at least B reachable states, so some state lies outside that ball.

Let A be the support of μ_{1000} . Then

$$|A| \leq 1800^{1000}, \quad \mu_{1000}(A) = 1. \quad (93)$$

Using the definition

$$\|\mu - \nu\|_{\text{TV}} = \max_E |\mu(E) - \nu(E)|, \quad (94)$$

we obtain

$$\begin{aligned} \|\mu_{1000} - \nu\|_{\text{TV}} &\geq 1 - \frac{|A|}{|\Omega|} \\ &\geq 1 - \frac{1800^{1000}}{B} \\ &> 1 - 10^{-148595}, \end{aligned} \quad (95)$$

where strictness follows from Equation~(91). No stationarity, independence, or uniform choice of letters is needed for this support argument.

These are statements about a specified alphabet and fixed-frame face-color states. Macro applications, simultaneous native commands, and camera motions are not single letters in this metric.

The distance bound is existential, not a lower bound for every scramble. A known 1000-letter scramble has an inverse solution of at most 1000 letters because $\mathcal{A}$ is inverse closed.

If a valid finite symmetry group L acts on the chosen state set, quotienting may reduce counts: generally

$$|\Omega/L| \geq \frac{|\Omega|}{|L|}, \quad (96)$$

rather than preserving $|\Omega|$. A quotient's goals and metric must also be specified before transferring distance claims. $\square$

2.15 Lemma 12: Geometric normalization of the legal group

Statement

Under R, F, and E, the geometric action of K normalizes G , its intersection with G is trivial, and

$$\langle G, K \rangle \cong G \rtimes K. \quad (97)$$

Proof

Let $k \in K$. Orthogonality and equivariance of poles give

$$k(U_c) = U_{kc}. \quad (98)$$

Let $R_{c,a}$ denote a legal cap rotation corresponding to $a \in A_4$: it rotates the cap and fixes its complement. On the complete subdivision,

$$kR_{c,a}k^{-1} = R_{kc,kak^{-1}}. \quad (99)$$

The restriction kak^{-1} fixes the new cap axis and is a proper rotation on its perpendicular three-space. This remains true when $\det k = -1$, because determinant is preserved by conjugation. It is consequently an element of the new tetrahedron's A_4 rotation group.

By F and Lemma 6, every such local rotation is a finite word in H_{kc}, T_{kc} . Therefore

$$kGk^{-1} \subseteq G. \quad (100)$$

Applying the same argument to k^{-1} gives equality.

Now suppose $g = k$ as sticker permutations, with $g \in G$ and $k \in K$. Every center slot is fixed by g , hence by k . Hypothesis E makes this mean

$$kn_c = n_c \quad \text{for every } c. \quad (101)$$

The poles span $\mathbb{R}^4$. Otherwise a nonzero vector orthogonal to every pole would give an unbounded line in the intersection of the facet half-spaces, contradicting boundedness of $\tilde{P}$. Thus k fixes a spanning set and is the identity. Since G is a subgroup of the sticker symmetric group, its action is faithful, and g is also the identity.

Normalization and trivial intersection imply that every element of $\langle G, K \rangle$ has a unique form gk . Multiplication is governed by the conjugation action of K on G , giving the asserted internal semidirect product.

The faithfulness requirements are essential. An action on a display quotient that identifies centers, or a geometric representation without a distinguished spanning center set, would not justify the intersection argument. $\square$

2.16 Distinct symmetry and motion groups

The groups appearing above serve different purposes.

Group	Action and role
$SO(4)$	Continuous orientation-preserving changes of four-dimensional viewing frame
$Sp(1) \times Sp(1)$	Double cover of $SO(4)$, with diagonal $\{\pm(1, 1)\}$ kernel
K^+	The 7200 orientation-preserving symmetries preserving the finite polytope
K	All 14400 geometric symmetries of the polytope
A_4 at one cap	Its twelve local tetrahedral rotations
G	The finite group of legal permutations of the retained sticker slots

To see the continuous covering directly, take $A \in SO(4)$ and put $p = A(1)$. The map $L_{p^{-1}} \circ A$ fixes 1 and restricts to an element of $SO(3)$ on the imaginary quaternions. Every such three-dimensional rotation is $x \mapsto qxq^{-1}$ for a unit quaternion q , so A has a unit-quaternion-pair representation. The kernel calculation is the same as in Lemma 4.

A generic camera rotation does not preserve the finite set V . A local cap turn generally is not a global rigid symmetry. Under C, the lower bound $B > 10^{151851}$ alone already separates the scale of the legal action from the finite geometric group.

2.17 Lemma 13: Conditions for sound symmetry reduction

Statement

Normalization of G by K does not imply preservation of the chosen 1800-symbol word metric.

A symmetry quotient preserves shortest-path distance to a goal set if the symmetries act by cost-preserving automorphisms of the actual allowed state graph, preserve the goal set and the required protection conditions, and quotient paths retain sufficient information to lift to actual labelled paths.

Proof

Normalization says that a conjugated generator is an element of G . It says nothing about its length in $\mathcal{A}$.

Already in one tetrahedral group, take

$$H = (12)(34), \quad T = (123). \quad (102)$$

Conjugating H by T gives a different half-turn. This is none of the three chosen local symbols H, T, T^{-1} , although it is a finite word in them. Thus the local generating set is not invariant under all local conjugations. Global geometric symmetry supplies no automatic remedy for this failure. A separate audit would be needed to show preservation of the particular global alphabet and its primitive costs.

Now let Γ be the actual allowed state graph, with edge costs, and let L act by cost-preserving graph automorphisms. Suppose the goal set is L -invariant. The graph must already encode the relevant allowed states, protected buffers or frames, legal generators, and any restrictions on intermediate behavior.

Projection sends every path to a quotient path of the same cost. Conversely, suppose an edge of a quotient path is represented by $u \rightarrow v$, while the current lifted state is $x = \ell u$. The edge lifts to

$$x = \ell u \longrightarrow \ell v \quad (103)$$

with the same cost, because ℓ is a graph automorphism. Repeating this lifts the entire path. If its quotient endpoint is a goal orbit, the lifted endpoint is a goal because the goal set is invariant. Therefore the quotient and original minimum costs agree.

To implement the lift, one must retain a representative edge, its actual move, and the symmetry needed to align its source with the current lifted state. Ordered frames, labels, and buffer identities must be transported as part of this information.

For fixed-frame solved colors, a geometric permutation of positions alone need not preserve the solved coloring or even the reachable color set: it moves centers with distinct colors. A combined position action and matching permutation of home-cell symbols may preserve solved, but that is a separately defined action.

Likewise, setwise preservation of two buffers is insufficient when the task protects them individually with ordered frames. The acting subgroup must preserve the actual requirement or carry explicitly tracked buffer data.

Without these conditions, a quotient may change the goal, admit forbidden transitions, reduce costs artificially, or lose the data needed to reconstruct a legal labelled solution. $\square$

2.18 Lemma 14: Complete endpoint transport and certified caching

Statement

Let W and S be verified legal sticker permutations, with chronological witnesses. Suppose the complete nonfixed map of W is

$$E_W = \{(u, W(u)) : W(u) \neq u\}. \quad (104)$$

Then the complete nonfixed map of the chronological conjugate $S^{-1}WS$ is

$$E_{S^{-1}WS} = \{(S(u), S(W(u))) : (u, W(u)) \in E_W\}. \quad (105)$$

Thus transporting both endpoints computes exactly the same full permutation as replaying the conjugated legal witness, including all collateral.

Proof

As an ordinary function, the chronological word $S^{-1}WS$ is

$$S \circ W \circ S^{-1}. \quad (106)$$

For each u ,

$$(S \circ W \circ S^{-1})(S(u)) = S(W(u)). \quad (107)$$

Since S is bijective, $S(W(u)) \neq S(u)$ exactly when $W(u) \neq u$. Every moved source of the conjugate is therefore represented, and no fixed source is included.

This proves equality on every sticker, not merely on a target orbit. Collateral cycles are transported by exactly the same formula.

Consequently a cached full map can be certified by a complete seed map, a verified setup witness, the endpoint transport calculation, and consistent permutation conventions. Cache identity must distinguish the underlying model, seed, and setup data sufficiently to avoid substituting a different map.

The result licenses exact reuse of a permutation. It gives no numerical performance guarantee and does not shorten the retained primitive witness. $\square$

2.19 Lemma 15: Guarded setup graphs and legal inverses

Statement

Fix two buffer positions with their sticker frames pointwise protected. On the finite graph of ordered target sticker frames, enlarging the positive guarded alphabet by legal inverses cannot increase any minimum unit-cost setup length.

For nonnegative unequal edge costs, minimum cost must be computed by a suitable weighted shortest-path algorithm rather than ordinary breadth-first search.

Proof

A vertex records a complete ordered target frame

$$f = (s_1, \dots, s_{k_o}) \quad (108)$$

at a nonbuffer position. An allowed primitive g induces the transition

$$f \mapsto (g(s_1), \dots, g(s_{k_o})). \quad (109)$$

Allowed moves fix both buffer sticker fibers pointwise. In the retained cap guard, forbidding every cap in

$$M(A) \cup M(B) \quad (110)$$

is a sufficient condition for this fixed-buffer property.

Let E_+ be the edges arising from allowed positive moves. Adding their legal inverses gives an edge set $E_\pm$ with

$$E_+ \subseteq E_\pm. \quad (111)$$

Every positive setup path is still available with the same unit cost. Thus, with infinite distance allowed for unreachable vertices,

$$d_\pm(f_0, f) \leq d_+(f_0, f). \quad (112)$$

In this setting inverse moves also fix the buffers. Moreover, $H_c^{-1} = H_c$ and $T_c^{-1} = T_c^2$, so inverse transitions already have positive legal witnesses. Adding inverse letters can shorten paths in the chosen alphabet without needing to enlarge its reachable component.

For unit costs, breadth-first search explores vertices in nondecreasing path length and finds minimum setup lengths in that graph. If costs are nonnegative but unequal, path length is no longer path cost. A weighted algorithm, such as Dijkstra's algorithm, is appropriate. Adding inverse edges still cannot increase minimum cost provided old edges retain their costs.

These statements compare precisely specified graphs. They do not claim strict improvement, numerical speedup, or global optimality among different buffer choices or setups allowed to move the buffers temporarily. $\square$

2.20 Scope tests: collateral, commutation, and quotient equality

Two elementary examples expose invalid stronger claims.

First, on six stickers the permutations

$$u = (1\ 2\ 3), \quad v = (1\ 2\ 3)(4\ 5\ 6) \quad (113)$$

have identical restrictions to the target orbit $\{1, 2, 3\}$ but different complete actions. Both are even. Equality in a target-orbit quotient therefore does not justify replacing a full-state macro. One must prove equality on the whole state, or explicitly justify the collateral difference under the application's current conditions.

Second, consider

$$a = (1\,2)(5\,6), \quad b = (3\,4)(6\,7). \quad (114)$$

If only stickers 1, 2, 3, 4 are displayed, their visible supports are disjoint. But their full actions do not commute: applying a then b sends 5 to 7, whereas applying b then a sends 5 to 6.

By contrast, permutations with disjoint full supports commute. Each permutation fixes the support of the other pointwise, and a permutation maps its own support to itself; checking a point in either support or outside both gives equality of the two compositions.

These examples refute the stronger quotient and visible-support claims. They are consistent with, and explain the necessity of, the complete-support hypotheses in Lemmas 9 and 14.

2.21 Theorem: Full 600-cell theory package

Statement

Let V, P, K, K^+ , the pole convention, and hypotheses R, F, I, C and E be as defined above. Then the following assertions hold, with the hypotheses indicated separately.

1. **Unconditional geometry and topology.** The radius-two polytope P is the regular $\{3, 3, 5\}$ 600-cell with edge length $2/\phi$, face vector $(120, 720, 1200, 600)$, tetrahedral facets and icosahedral vertex links. Five facets meet at each edge and twenty at each vertex. Its boundary is radially homeomorphic to S^3 , with integral homology $\mathbb{Z}$ in degrees zero and three, zero in degrees one and two, and trivial fundamental group. Its polar is a regular 120-cell. The finite checker verifies its specified arithmetic and incidence statements; mod-2 Betti numbers alone do not establish this topological identification.
2. **Unconditional geometric symmetry.** The unit vertices $V/2$ form I^* , and the left/right quaternion action identifies K^+ with $(I^* \times I^*)/\langle(-1, -1)\rangle$, of order 7,200. Quaternion conjugation extends this action to $K = W(H_4)$ of order 14,400, with the Coxeter presentation stated in Lemma 4. These finite groups are distinct from $SO(4)$, the rotation group of a cap and the legal twist group.
3. **Retained cut model, under R and F.** The 600 oriented poles describe 300 antipodal axes with two outer layers at normalized distance 121/125 and a fixed middle layer during an outer-layer turn about that axis. The retained model has 259,800 sticker slots, 177,120 surface pieces, 600 fixed centers and the stated 35 moving orbits. Sticker-to-piece incidence is equivariant under legal twists. Complete cap signatures and hosting sets provide the specified addresses, while $|M| - 1$ is a membership rank rather than a dimension. The pair H_c, T_c generates the full local A_4 action. The count of 433 sticker regions per facet remains a retained-model input.
4. **Legal actions and conditional bounds.** Under R and F, the legal group embeds in the stated product of transported-frame wreath products, with chronological multiplication $(p, a)(r, b) = (r \circ p, c)$ and $c_i = a_i b_{p(i)}$. With the additional generator-invariant hypothesis I, positional parity and orientation abelianization give

$$|G| \leq \frac{\prod_o N_o! |H_o|^{N_o}}{2^{4352}}. \quad (115)$$

With the separate pure-controller hypothesis C, the supported action contains $A_{3600} \times A_{2400} \times A_{7200}^7$ and yields the balanced fixed-frame color lower bound B of Lemma 10. Neither bound determines the exact order of G or proves sufficiency of the listed invariants.

5. **Specified word metric, under R and C.** For the 1,800-symbol alphabet $\{H_c, T_c, T_c^{-1}\}$, the word-ball inequality implies a worst-case distance of at least 46,648. Any 1,000-step distribution from solved has total-variation distance strictly greater than $1 - 10^{-148595}$ from uniform reachable fixed-frame face-color states. This is not a lower bound for each scramble; a recorded 1,000-letter word has an inverse of at most 1,000 letters. Labelled states, face-color states and symmetry quotients remain distinct.

6. **Extended geometric action, under R, F and E.** The faithful geometric action on the complete retained subdivision normalizes G . Since legal moves fix every distinguished center and those centers span the geometric space, $G \cap K = \{1\}$ and $\langle G, K \rangle \cong G \rtimes K$. Normalization does not automatically preserve the chosen move alphabet or cost. Sound shortest-path reduction requires the graph, goal, protection and reconstruction conditions of Lemma 13.
7. **Sound optimization.** Under the complete-action and guarded-graph hypotheses of Lemmas 14 and 15, transport of both support endpoints computes the full conjugated net permutation, and adding legal inverse edges cannot increase minimum unit-cost setup length. Unequal nonnegative costs require weighted shortest-path search. Equality on a target orbit does not establish equality of full-state macros; disjoint full supports suffice for commutation, while disjoint visible supports do not. No numerical speedup, new solve or performance record follows from these statements.

Proof

Lemmas 1–3 establish the unconditional coordinate geometry, face incidences, radial homeomorphism, integral homology, fundamental group, and polar duality required in part (1). The finite-checker discussion specifies the exact enumeration algorithm, the supporting-facet completeness argument, and the conclusions outside its computational scope.

Lemma 4 identifies the quaternion symmetry action, proves its kernel, establishes both the lower and upper symmetry counts, and proves the Coxeter presentation required in part (2). The continuous covering argument and the group-role table distinguish camera motions, finite geometric symmetries, local cap rotations, and legal sticker permutations.

Lemma 5 gives the exact scale conversion for part (3). The retained-model definitions specify the 600 oriented caps, their 300 antipodal axes, signatures, hosting sets, and census without claiming an independent cut reconstruction. Lemma 6 proves local A_4 generation and piece equivariance and explains the application rank.

Lemma 7 proves the faithful ambient embedding and chronological frame law. Lemma 8 derives the conditional invariant upper bound under I, including independence of its ambient index factors. Lemmas 9 and 10 derive the independent alternating actions and balanced-color lower bound under the separate pure-controller hypothesis C. These establish part (4) without asserting full reachability or an exact legal-group order.

Lemma 11 proves part (5) from the word count, the color lower bound, and explicit integer inequalities. It distinguishes worst-case existence from individual scramble length and distinguishes fixed-frame color states from labelled and symmetry-quotiented states.

Lemma 12 proves part (6)’s normalization, trivial intersection, and semidirect product under the complete geometric realization E. Its proof uses the faithful sticker action and the spanning geometric center set explicitly. Lemma 13 shows why normalization alone is insufficient for the chosen metric and proves the conditions and reconstruction obligation for sound shortest-path symmetry reduction.

Finally, Lemmas 14 and 15 prove the two optimization statements in part (7). The complete-support examples establish the stated limitations on macro substitution and commutation. None of these arguments asserts a numerical speedup, a new solve or performance record, or a determination of the unknown full legal-group order. $\square$

3 From theory to implementation

The preceding results offer several ways to reduce work without changing the puzzle. The practical target is preservation of the full labelled action. A faster operation is useful only if it carries the same piece identities, orientations, collateral effects and history semantics as its reference implementation.

3.1 Normalization without changing model identity

For the radius-two coordinates, the supporting-facet calculation gives squared inradius

$$r^2 = \frac{(4 + 6\phi)^2}{16 + 24\phi} = 1 + \frac{3}{2}\phi = \frac{\phi^4}{2}. \quad (116)$$

The retained pole convention has $\|n_0\|^2 = \phi^6 + 3 = 8\phi^2$. Thus the ratio between its inradius and the radius-two model's inradius is $4/\phi$. A suitable orthogonal identification and that scale relate the two geometric descriptions. They do not determine the application's cell numbering or sticker IDs. The normalized cap plane at 121/125 of the cell distance lies inward by 4/125 of the inradius. This is a geometric distance ratio, not a statement that 3.2 percent of the pieces move.

Keeping the normalization explicit prevents a particularly subtle error: applying the numeric constant 0.968 to coordinates whose supporting planes have a different distance from the origin. Any replacement geometry pipeline should transform both positions and plane equations coherently, then verify its cell and sticker correspondence against the immutable model manifest.

3.2 State representation and incremental updates

Let S be the fixed set of 259,800 sticker slots. Store $L[s]$ as the identity of the sticker currently occupying slot s . A legal permutation $m : S \rightarrow S$ is applied by

$$L_{\text{new}}[m(s)] = L_{\text{old}}[s]. \quad (117)$$

Let $D = \text{supp}(m)$, and let $\pi : S \rightarrow \mathcal{P}$ map a slot to its host piece position. Since a permutation preserves its own support, only the destination positions in $\pi(D)$ can have changed their occupants or ordered frames. This licenses an incremental progress update: update the inverse location map and recalculate correctness for the affected piece positions, then adjust the per-orbit totals. It does not license dropping stickers from the mechanical state.

The present `PuzzleState.refresh` reconstructs location and correctness information with full-array work. A future incremental implementation should retain the full refresh as an independent oracle. For each legal generator, its inverse, a composite macro and an imported replay, compare the complete inverse-location arrays, every piece-correctness bit and every per-orbit count. A discrepancy must reject the optimization before it can affect a saved session. Dense moves may make a full vectorized refresh faster than bookkeeping over a large support; the decision should be measured against support size.

The model identity and the full-state hash have different purposes. The former fixes geometry, IDs, cuts, frames and generator meaning; the latter identifies a particular labelled configuration. Neither may be replaced by a smaller displayed subset. In particular, an incremental noncryptographic XOR digest is not a substitute for the current SHA-256 boundary used by previews and certificates. A change to hashing semantics would require an explicit protocol version and new validation.

3.3 Certified net maps and caches

A long legal word may have a small net support. The endpoint-transport result in the blueprint justifies caching the complete conjugated net permutation while retaining its primitive witness. Applying that

cache can reduce execution work from repeated full-word replay to movement of the net support. This is a distinction between execution cost and mathematical word length: it does not shorten the recorded legal witness.

A cache record should bind at least the model identifier, frame convention, seed identifier and digest, setup word and direction, complete net-map digest, and protected-orbit contract. Retain both source and destination endpoints and verify bijectivity. An orbit-only map is inadequate for a raw seed whose other effects occur in later stages. Two macros with the same target three-cycle may be incompatible substitutes because they move different collateral pieces.

There are also two separate symmetry-caching opportunities. For geometric data, an exactly verified element of K can transport incidence and sticker endpoints. For search, the same element is useful only if it preserves the allowed transitions, edge costs, distinguished buffers and goal condition, or if the search explicitly transforms all of them. The theorem $K \leq N(G)$ concerns legal reachability; it does not establish that K acts by isometries of the chosen 1,800-symbol metric.

3.4 Search graphs, orientation and cost

The guarded setup graph tracks an **ordered target sticker frame**, with both buffers fixed. Forgetting frame order merges distinct orientations. That merge can yield a setup that places the correct piece in the correct position while leaving a nonabelian orientation error. The retained controller graph has at most

$$V_o = (N_o - 2)|H_o| \quad (118)$$

candidate target frames for orbit o . A transition that applies a guarded generator must transport the whole frame. With b generators and k sticker entries per frame, straightforward graph construction costs $O(V_o bk)$, apart from generator storage. A predecessor tree needs $O(V_o)$ records rather than storing a complete setup word at every vertex.

Adding legal inverses to a positive unit-cost alphabet cannot increase any shortest setup distance. It may nevertheless increase the time spent building the graph, so shorter words and lower build latency should be measured separately. Ordinary BFS is appropriate for unit costs. Primitive length, elapsed execution cost and ergonomic cost are different nonnegative weight functions; a weighted search must use an appropriate algorithm such as Dijkstra’s algorithm. Bidirectional search needs correctly reversed transitions and compatible fixed-buffer conditions.

The retained workload tables identify O02, O15 and O29 as expensive targets: together they account for about 40.34 percent of the constructive primitive bound while requiring about 3.8 percent of its stars. Shorter complete witnesses for those orbits could therefore improve the primitive bound disproportionately. They need not improve interactive speed by the same amount when the workbench already executes cached net permutations. Every replacement requires full-support replay and revalidation of the collateral order.

3.5 Stable rendering through finite symmetry

The geometric relationship between cells suggests an instancing design. Let v_j be vertices of one reference cell’s subdivided geometry and let R_c place that geometry in cell c . Let $Q \in SO(4)$ be a camera rotation. A candidate rendering pipeline computes

$$v_{c,j} = R_c v_j, \quad u_{c,j} = Q v_{c,j}, \quad p_{c,j} = \Pi_3(\Pi_4(u_{c,j})). \quad (119)$$

Here Π_4 is a specified 4D-to-3D projection and Π_3 the final screen projection. For example, a perspective projection may use $\Pi_4(x, y, z, w) = d(x, y, z)/(d - w)$, restricted away from its singular plane. Clipping must handle the valid domain before division; a small denominator is not a reason to corrupt coordinates

or alter the puzzle state. The real application may use additional shrink, visibility and view parameters, which must be reproduced as well.

The web mesh asset has 30,480 base vertices and 10,160 base triangles. Replicating that mesh over 600 cells represents 6,096,000 triangles before view-dependent rejection. This is a count for the web asset, **not a measurement of the native DirectX renderer’s mesh**. Sticker count alone is therefore an inadequate graphics workload descriptor. Profiling should report actual submitted vertices, triangles, lines, draw calls and uploaded bytes for the native path.

One prospective GPU implementation would retain immutable reference geometry and cell transforms, update the camera in a small uniform buffer, and obtain colors and selection styles through a sticker-ID lookup. Such a design could reduce repeated CPU projection and packing. It still has to reproduce clipping, ordering, transparency and selection. The installed Managed DirectX path and device capabilities constrain what is practical; a renderer rewrite is a separate, tested engineering change, not a consequence already implemented by the group theory.

The existing native optimization reuses a 28-byte-per-vertex upload array around a pinned upstream renderer. This reduces allocation and marshaling while preserving the established projection path. It leaves CPU projection and triangle/line packing in place. A useful next experiment must therefore time those stages separately before attributing the remaining delay to the GPU.

3.6 Filtering, framework visibility and picking

A filter is a predicate on pieces, potentially using their solved identity, current position, orbit, rank, correctness or an explicit context rule. It determines which geometry is displayed; it does not define a smaller puzzle. All stickers belonging to a hidden piece must be excluded consistently from rendering and hit testing. Buffer or preview context outside the selected predicate should be explicit, so additional pieces cannot be confused with a failed filter.

The 600-cell framework is a separate draw layer. Hiding it must suppress its lines at every draw entry point, including animation and direct redraws. It must retain the cell planes and clipping information used by visible stickers. Removing structural clipping data to remove a visual outline conflates geometry with presentation and can produce both rendering and picking defects.

Render and pick operations should consume a coherent snapshot:

$$\mathcal{R} = (\text{revision}, L, Q, F, \text{frame policy}, \text{motion policy}). \quad (120)$$

The picker’s candidate set must use the same visibility predicate F as the displayed scene. A cached projection is valid only for the matching revision, camera and view parameters. After a sampled motion frame, exact picking needs either a refreshed projection of the complete visible geometry or a mathematically equivalent ray test against that geometry. Invisible triangles and stale axes must not remain interactive.

During movement, adaptive detail can improve responsiveness while the complete state remains unchanged. When movement stops, restoring all visible geometry is a distinct latency that should be reported. A sampled frame must never be counted as a successful full-detail frame in the 30 FPS qualification workload.

3.7 Transactions and optimization boundaries

The state transition remains authoritative. A preview binds the current revision, full-state hash and protection constraints; commit rechecks all three. The complete net permutation is applied and journalled atomically before the new head is exposed. Rendering consumes the resulting immutable snapshot. Changing frame visibility, camera pose or sampling policy must leave the state hash unchanged.

Reset and import require complete validation before durable writes and a recoverable previous state. Checkpoints and log exports should record the model identity, chronological moves or certified macros, and enough state information to verify replay. A failed image update after a successful database commit

must not cause the operation to be applied twice. These contracts matter especially for scrambling: color transport follows the committed labelled permutation rather than a partly updated display array.

The rendering queue may coalesce obsolete camera snapshots when a newer view is available. It may not coalesce or discard committed mechanical moves from the history. Likewise, progress reports may cache aggregates, but sessions reports must describe durable operations rather than estimates derived from animation completion.

3.8 What performance evidence supports

The frozen 0.2.4 validation recorded an optimized full-detail mean of 453.45 ms and a 95th percentile of 533.82 ms at a 931×604 viewport. The comparison path averaged 626.57 ms. The 27.63 percent reduction comes from 18 fixed-pose redraw measurements, not a sustained rotation test. The reciprocal of 453.45 ms is approximately 2.2 frames per second; it is not a qualified 1080p frame rate.

For a mean baseline T and a measured nonaccelerated fraction f , an idealized Amdahl model gives

$$T_{\text{new}} = fT + \frac{(1-f)T}{r}. \quad (121)$$

Even infinite acceleration of the second term could reduce 453.45 ms to 33.33 ms only if $f < 0.0736$. The prior total process-CPU figure is not a measurement of this critical-path fraction; CPU and GPU overlap, waiting, packing and submission must be separated before using the model quantitatively. More GPU throughput alone cannot be promised to satisfy the target.

The target minimum remains **30 FPS at 1920 by 1080 with all 259,800 stickers present at full detail**. No specific minimum GPU model is certified. Intel HD Graphics 620 supports the measured filtered/adaptive workflow but has not passed this full-detail qualification. Neither the new theory nor the exact geometry audit changes that status.

Proposed experiment	Evidence required before acceptance
Incremental state/progress updates	Complete state, inverse-location map and orbit counts agree with full recomputation
Symmetry or macro cache	Complete labelled net map equals legal witness, including all collateral
Inverse or weighted setup search	Correct ordered frame, both buffers fixed, measured cost and reconstructed legal word
GPU projection or instancing	Clipping and reference-image agreement, stable sticker IDs and matching picking
Full-detail rotation qualification	At least 60 seconds each of warmed 3D and 4D motion at 1080p; full workload; frame-time distribution and presentation method
Adaptive motion	Interaction latency, exact restoration latency, unchanged mechanical hash and correct post-motion picking

For the full-detail qualification, report mean, median, 95th and 99th percentile frame times; target a 95th percentile no greater than 33.33 ms as well as sustained 30 FPS. Identify the application build, driver, viewport, clipping settings and actual visibility count. Check representative images against the rendering reference, and verify identical labelled hashes before and after camera-only runs. Cold construction, restore latency and device recovery are separate measurements. These are acceptance criteria for future work, not claims that the present supplement ran those graphics experiments.

4 Reproduction and interpretation

The accompanying standard-library geometry audit is self-contained. It constructs the 120 exact vertices, enumerates adjacency and simplex cliques, verifies supporting facets, forms simplicial boundary maps over $\mathbb{F}_2$, computes graph distances, and enumerates the quaternion-induced symmetry maps. Its JSON result records both numerical outputs and the scope boundary. Running it does not open the application or read a user session.

The original 35-orbit algorithms paper and the curated replay archive remain unchanged. They supply the complete algorithm cards and finite words. The existing configuration-bound and extended-derivation audits remain the references for retained controller and arithmetic evidence. The new checker addresses base geometry; it does not replace those audits.

Rethlas was used at upstream revision `887cc46427636bbdd235160a112f9a30ae81d040`, with separate generation and verification invocations of Astra. The stock file-based runner could not read its task inputs under the nested Windows execution policy. The adaptation supplies the reviewed public inputs as text to read-only agents and lets the external coordinator persist outputs through Rethlas’s memory and strict verification schema. The coordinator retains rejected candidates and repair reports; acceptance requires the verifier’s error and gap lists both to be empty. Public records contain the reviewed blueprint, exact verdicts, scope and hashes, while local paths, raw execution logs and unrelated material are excluded.

The first complete inline blueprint, comprising fifteen lemmas and the combined theorem, received the strict verdict **correct**, with zero critical errors and zero gaps. No mathematical repair round was required. Earlier file-policy attempts produced no candidate and are not counted as completed reviews. The exact returned blueprint and verdict are retained separately. This typeset exposition removes the generation-time status paragraph, normalizes equation references, and replaces imperative task wording in the combined statement with an equivalent declarative statement. The engineering discussion is a separately authored analysis.

The resulting mathematical review is informal. It neither proves the exact legal-group order nor formally verifies the entire software stack. Source provenance, exact finite checks and explicit hypotheses make its conclusions inspectable; independent reproduction and mathematical review remain possible without the private design transcript.

5 Reading guide

1. **Hypercubing community, Formal Introduction to Grip Theory.** Defines active grips, attitudes and the distinction between general piece states and twist-reachable states. Read it alongside the blueprint’s frame convention; the program’s finite sticker action need not retain an invisible geometric attitude. <https://hypercubing.xyz/theory/grip-theory/formal/>
2. **Hypercubing community, Piece Invariants.** Background on parity and orientation constraints. Its hypercube monoflip example is illustrative rather than a full-600-cell certificate. <https://hypercubing.xyz/theory/invariants/>
3. **Hypercubing community, God’s Number.** Explains metric-specific counting arguments and the distinction between a worst-case bound and a solution length for one state. <https://hypercubing.xyz/theory/gods-number/>
4. **Hypercubing community, Cut Depths.** Establishes normalized cut-distance conventions and illustrates how changing cuts changes a puzzle’s pieces. It does not enumerate the retained full-600-cell subdivision. <https://hypercubing.xyz/theory/cut-depths/>
5. **John C. Baez, From the Icosahedron to E8.** Primary background for binary icosahedral quaternions and 600-cell coordinates; arXiv:1712.06436v2. <https://arxiv.org/abs/1712.06436>
6. **Jihyun Choi and Jae-Hyoun Lee, Binary Icosahedral Group and 600-Cell.** *Symmetry* 10(8), 326 (2018). A primary geometric treatment connecting quaternion multiplication to the regular polytope. <https://doi.org/10.3390/sym10080326>
7. **David Vogan, Regular polyhedra and Coxeter groups.** Tufts lecture, 25 January 2019. Background for flag counting and finite reflection groups. <https://math.mit.edu/~dav/regpolyTUFTSHO2.pdf>
8. **Rethlas.** Upstream generation and independent proof-verification workflow. The companion provenance describes the local I/O adaptation and records the actual review outcome. <https://github.com/frenzymath/Rethlas>
9. **C600 Studio research materials.** The technical report, unchanged original algorithms paper, synthetic replay supplement, audits, and Full Detail Rotation requirements. <https://github.com/KonomiYuzu01/C600-MPUlt/tree/main/research>
10. **Andrey Astrelin, Magic Puzzle Ultimate.** Original simulator and renderer underlying this mod. <https://superliminal.com/andrey/mpu/>

Web references were checked on 7 September 2026. These links identify public technical sources, not personal conversation histories.